

AGENTIC AI & AUTOMATION

Professional Program

Advanced professional training in autonomous AI agents, LLM applications, orchestration, automation, and production deployment.

IN COLLABORATION WITH



6 Months

Live Online + Recordings

12 Modules

Structured Learning Path

10+

Hands-on Projects

1

End-to-End Capstone

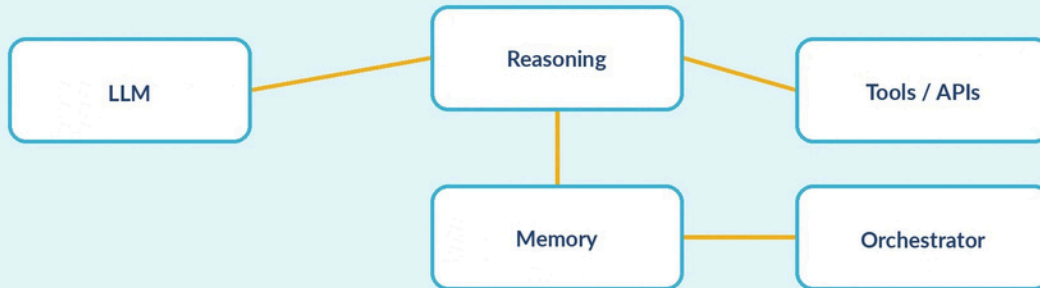
Python • LLM APIs • LangChain • LangGraph • CrewAI • AutoGen • RAG • FastAPI • Docker • Azure

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About the Course

A practical, code-first path from LLM foundations to autonomous AI systems.

Agent architecture: reasoning + tools + memory + orchestration



About Microsoft and Sky States Collaboration

The original Sky States brochure positions the program around collaboration with Microsoft, industry-relevant learning, hands-on technology exposure, and career-oriented preparation. For this new program, that same Sky States learning philosophy is applied to Agentic AI & Automation: learn the architecture, build working systems, evaluate them, and take them toward production.

1 Industry-oriented

Practical modules built around current agent frameworks, APIs, automation patterns, and deployment.

2 Hands-on learning

Every stage produces an implementation: agents, RAG pipelines, workflows, multi-agent systems, and deployment.

3 Career readiness

Portfolio-focused projects, architecture thinking, technical communication, and interview-oriented preparation.

WHY AGENTIC AI & AUTOMATION MATTERS

Why Agentic AI Matters

AI is moving from systems that answer questions to systems that can plan, use tools, and execute workflows.

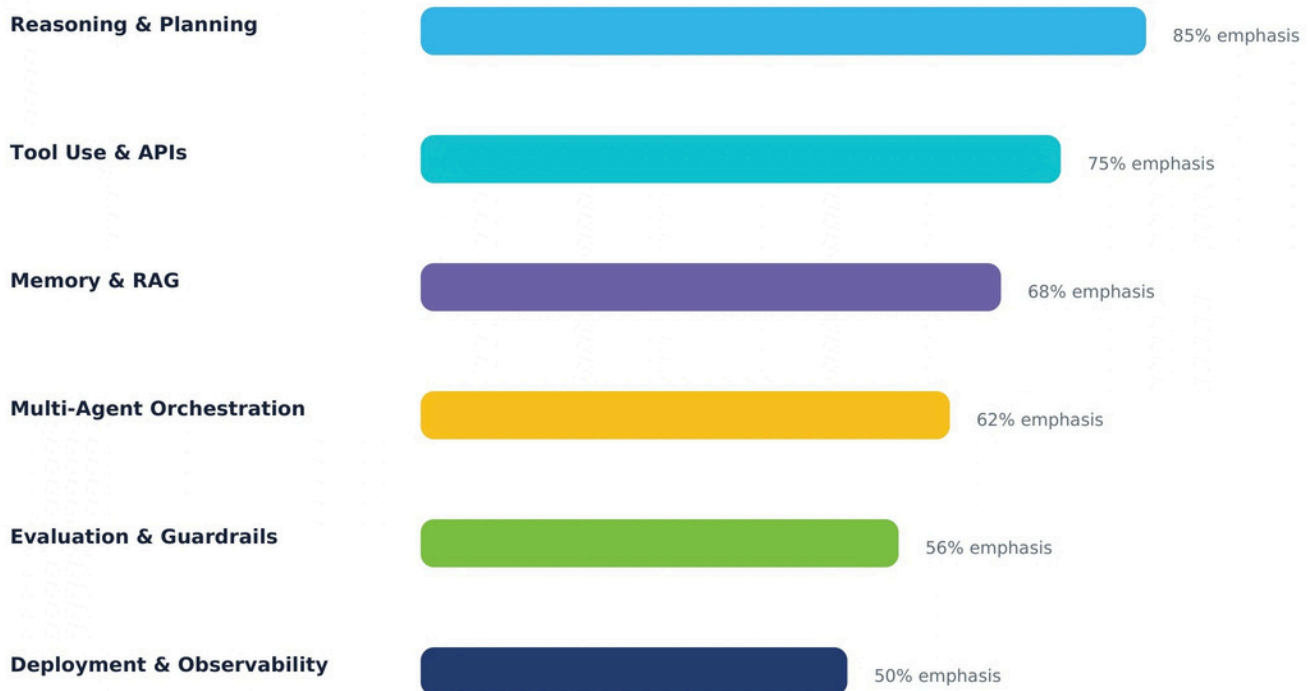
A
From assistant to actor

Modern agentic systems combine LLM reasoning with tools, memory, APIs, and orchestration so software can pursue multi-step goals.

B
Enterprise automation

Agentic workflows can support research, software engineering, data analysis, customer operations, document workflows, and internal automation.

Agent capability stack



Where agents can create value

Research & knowledge work

Data analysis

Workflow automation

Document intelligence

Software engineering

Customer operations

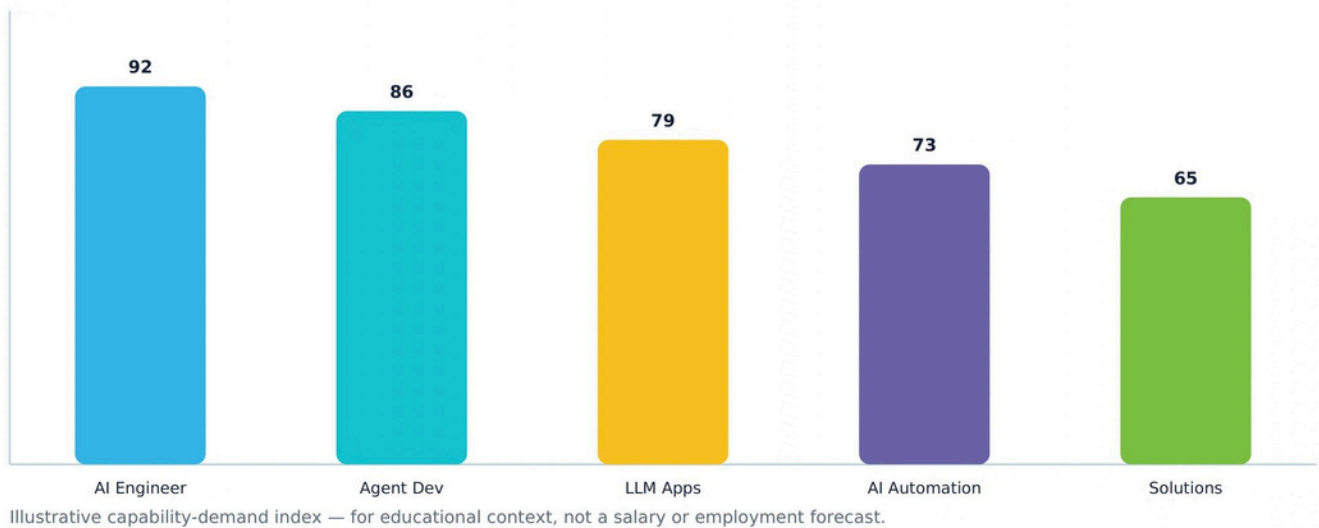
DevOps & IT operations

Multi-agent coordination

CURRENT TRENDS IN AGENTIC AI JOB MARKET

Current Trends in Agentic AI & Automation

The market is increasingly rewarding practitioners who can move from LLM experimentation to reliable systems.



BATCH PROFILE

Designed for working professionals and practitioners from software, data, ML, automation, DevOps, and adjacent technical backgrounds.

Best suited for

Software engineers, data scientists, ML practitioners, automation/DevOps engineers, AI product professionals, and career switchers with Python experience.

Prerequisites

Python programming experience is recommended. Familiarity with basic ML/AI concepts is helpful; the program builds the agent-specific foundations step by step.

PROGRAM HIGHLIGHTS

Industry-relevant skills for the agentic era

A six-month learning path designed around implementation, frameworks, projects, and production practices.

Agent Architecture

Reasoning, planning, memory, tools, and orchestration

01 6-month structured path

12 modules, 10+ hands-on projects, and one end-to-end capstone.

Framework-Native Skills

LangChain, LangGraph, CrewAI, AutoGen

02 Project portfolio

Working agents, RAG, LangGraph, multi-agent, code execution, and deployment deliverables.

Production Engineering

FastAPI, Docker, monitoring, tracing, cloud

03 Career-oriented delivery

Industry-facing architecture, project documentation, technical communication, and career support.

Live + recordings

Structured sessions with resources for review.

Industry-grade capstone

A complete autonomous agent system.

Practical assessments

Implementation-focused learning checkpoints.

Microsoft-aligned exposure

Relevant Microsoft cloud/certification context where applicable.

PROGRAM BENEFITS

1 Familiarity with essential tools

Python, LLM APIs, LangChain, LangGraph, CrewAI, AutoGen, RAG, FastAPI, Docker, LangSmith and cloud deployment.

2 Certificate / Credentials

Program completion recognition and relevant certification preparation where supported by the Sky States offering.

3 Curriculum

Core curriculum delivered through live online learning and structured self-paced resources.

4 Practical Knowledge

Implementation of agents, tool calling, memory, RAG, planning, multi-agent systems, evaluation, and deployment.

5 Hands-on Projects

10+ project-based deliverables across agent architecture, retrieval, orchestration, code execution, and production.

6 Support & Guidance

Learning resources, doubt support, structured feedback, and career-oriented guidance.

7 Personalised Attention

Small-group/mentor-led learning and focused project feedback.

8 Career Guidance

Role mapping, resume/project presentation guidance, interview preparation, and portfolio readiness.

Data & Agent Engineering with LLMs

Module 01 — Foundations of Large Language Models

- LLM fundamentals
- Transformers & attention
- Tokenization & context windows
- LLM APIs & model interaction
- Prompt fundamentals
- Structured outputs

Module 02 — Introduction to AI Agents

- Agent architecture
- Reasoning & planning
- Agent loops
- ReAct architecture
- Tool-aware agents
- Autonomous workflows

Module 03 — Tool Use & Function Calling

- Tool calling
- Function calling
- API integration
- Tool schemas & selection
- Execution & error handling
- Automation workflows

Module 04 — Memory Systems for Agents

- Short-term memory
- Long-term memory
- Conversation memory
- Context management
- Persistent state
- Retrieval-based memory

Frameworks & Stateful Agent Systems

Module 05 — LangChain for Agent Development

- Chains & runnables
- Prompts & models
- Tools & agents
- Retrievers
- RAG
- FAISS / Chroma

Module 06 — LangGraph — Stateful Agent Workflows

- Nodes & edges
- State management
- Conditional routing
- Human-in-the-loop
- Error recovery
- Multi-step orchestration

Module 07 — Multi-Agent Systems: CrewAI & AutoGen

- Agent roles
- Task delegation
- Agent communication
- Sequential workflows
- Parallel workflows
- Collaborative problem solving

Module 08 — Planning & Task Decomposition

- Planning agents
- Task decomposition
- Goal-oriented reasoning
- Hierarchical planning
- Dependency management
- Re-planning

Advanced Agents, Safety & Production

Module 09 — Code Generation & Execution Agents

- Code generation
- Code interpretation
- Sandboxed execution
- Automated debugging
- Testing
- Execution safety

Module 10 — Evaluation, Safety & Guardrails

- Agent evaluation
- Output validation
- Guardrails
- Hallucination mitigation
- Prompt injection awareness
- Human approval workflows

Module 11 — Production Deployment

- FastAPI & REST APIs
- Docker & containers
- Environment & secrets
- Cloud deployment
- Monitoring & logging
- LangSmith tracing

Module 12 — Capstone: Autonomous Agent Project

- Problem definition
- Solution architecture
- Tool integration
- Memory & RAG
- Evaluation & guardrails
- Deployment & final demo

PROJECTS & LEARNING PATH

Build, not just learn

Every stage of the program produces a working deliverable that can become part of a professional portfolio.

01 ReAct Agent from Scratch

Reasoning-action agent implemented without relying entirely on an orchestration framework.

02 RAG Research Pipeline

Document retrieval, grounded generation, and evaluation workflow.

03 LangGraph Workflow Agent

Stateful multi-step pipeline with branching logic and recovery.

04 Multi-Agent Crew

Collaborative CrewAI/AutoGen system for research and drafting.

05 Code Execution Agent

Agent that generates, runs, tests, and debugs Python safely.

06 Production Deployment

FastAPI-served agent with Docker and observability.

Your 6-month learning journey

M1-2

Foundations

LLMs, agents, tools, memory

M2-4

Frameworks

LangChain, LangGraph, multi-agent

M4-5

Advanced

Planning, code, safety

M5-6

Deployment

Production + capstone

More real-world agent scenarios

Practical agent concepts mapped to realistic business workflows and technical implementations.

1

Knowledge Base Copilot

RAG + memory + tools for enterprise knowledge workflows.

2

Recruiting Workflow Agent

Resume analysis, role matching, outreach drafting, and structured workflow handoffs.

3

Financial Research Agent

Research, retrieval, calculation tools, validation, and analyst-ready reporting.

4

Healthcare Operations Agent

Document and workflow assistance with strict human approval and safety controls.

5

Sales Intelligence Agent

Research accounts, summarize signals, prepare briefs, and update approved systems.

6

Software Engineering Agent

Issue analysis, code generation, testing, debugging, and pull-request assistance.

CAPSTONE PROJECTS

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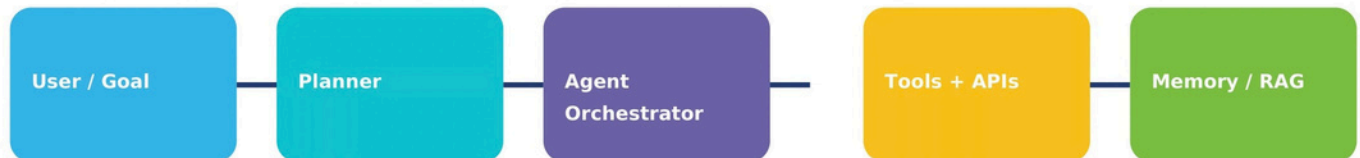
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06 Software Engineering Agent

Issue analysis, code generation, testing, debugging, and pull-request assistance.

An End-to-End Autonomous AI System

Design, build, evaluate, deploy, and present a production-oriented agentic solution.



The capstone demonstrates the complete agent lifecycle: goal → planning → tool use → state/memory → evaluation → deployment.

Architecture

System diagram, component responsibilities, data flow, and failure modes.

Implementation

Working agentic system using relevant frameworks and APIs.

Evaluation

Test cases, quality checks, guardrails, and reliability criteria.

Deployment

Containerized service, endpoint, monitoring/tracing, and operational notes.

Documentation

Professional README, setup instructions, architecture notes, and demo.

Review

Mentor-reviewed presentation and technical walkthrough.

TOOLS COVERED

The agent technology stack

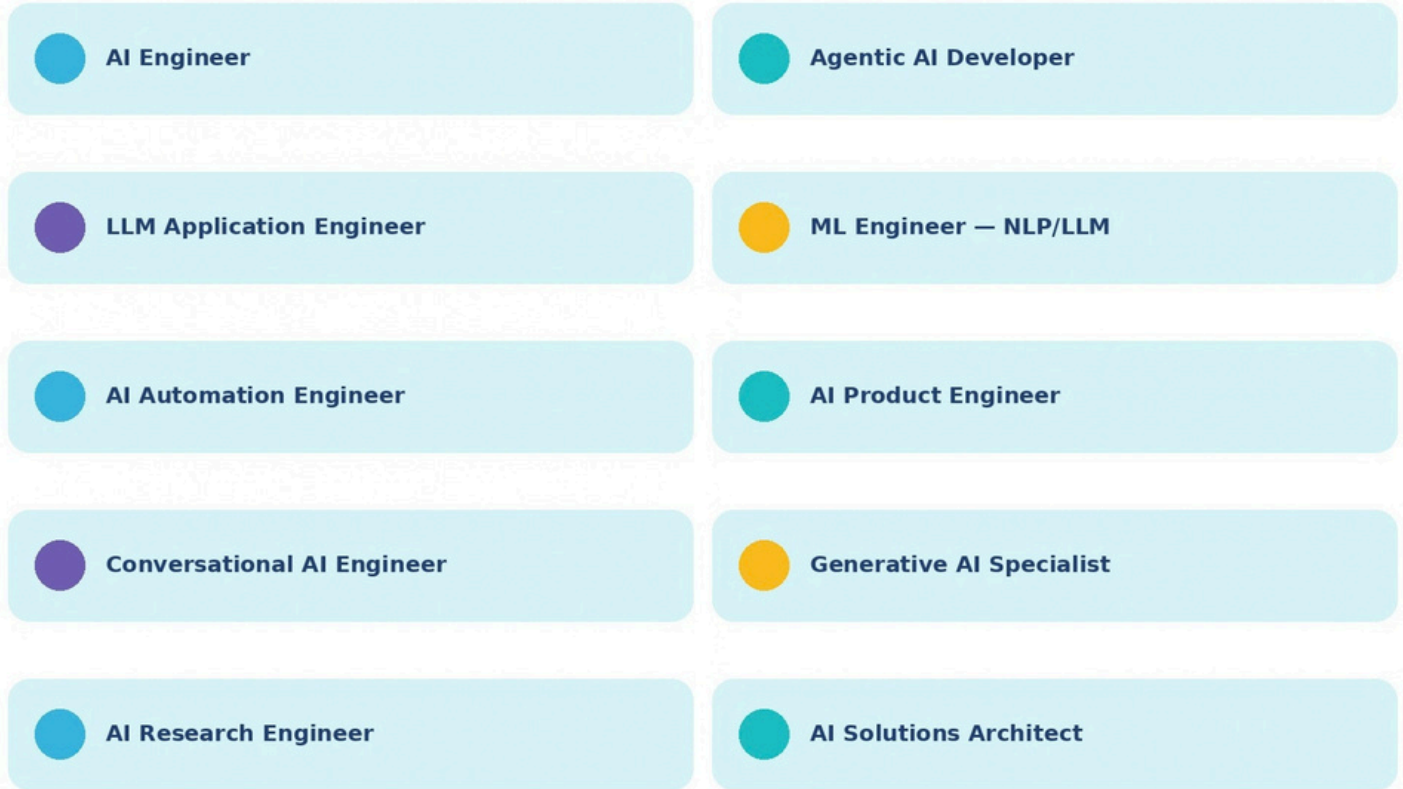


Framework-native learning

Build with the same categories of components used in production agentic systems: models, tools, state, retrieval, orchestration, serving, and observability.

From program to professional capability

The program maps agentic engineering skills to roles across software, data, ML, automation, and AI product teams.



Career readiness support

- Resume positioning for AI/LLM roles
- LinkedIn and professional profile guidance
- GitHub portfolio and README review
- Technical interview drills
- Agent architecture / system-design practice
- Capstone presentation coaching
- Role mapping by prior background

Build capability you can demonstrate

The program is designed to leave you with both knowledge and tangible evidence of implementation.

1 Foundations

Understand LLMs, transformers, prompting, structured outputs, and agent architecture.

2 Production Skills

Serve agents with FastAPI, package with Docker, and introduce monitoring/tracing.

3 Portfolio

Show working RAG, LangGraph, multi-agent, code-execution, and deployment projects.

4 Safety

Understand evaluation, guardrails, prompt-injection awareness, and human approval patterns.

5 Framework Depth

Use LangChain, LangGraph, CrewAI, and AutoGen in realistic workflows.

6 Career Readiness

Translate project work into resumes, GitHub artifacts, technical demos, and interview stories.

Outcome

Build, evaluate, deploy, document, and explain agentic systems with practical evidence of implementation.

Completion & Program Details

A professional learning record supported by structured modules, projects, and a capstone.

CERTIFICATE OF COMPLETION

Certified Agentic AI & Automation Professional

Awarded by Sky States upon successful completion of the program

6-Month Structured Program

Program	Agentic AI & Automation Professional Program
Duration	6 Months
Mode	Live Online + Recordings
Level	Intermediate-Advanced
Curriculum	12 Modules • 10+ Hands-on Projects
Capstone	1 End-to-End Autonomous Agent Project
Certificate	Certificate of Completion

BUILD THE FUTURE WITH AGENTIC AI

Design, build, and deploy autonomous AI systems.

A structured Sky States professional program for practitioners ready to move from LLM experimentation to reliable agentic workflows and production delivery.

SPEAK TO AN ADVISOR

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